



Dataset Preparation & Upload Guidelines

To ensure accurate and reliable **DDoS detection results**, your dataset must follow a specific structure. Please read the guidelines below carefully before uploading your data.

1. Creating Your Own Dataset Using CICFlowMeter

To create a custom network traffic dataset, you should use **CICFlowMeter**.

CICFlowMeter is an **official and widely used traffic flow generation tool** developed by the Canadian Institute for Cybersecurity. It converts raw network traffic into flow-based features that are internationally recognized and commonly used in research datasets such as CICIDS and CIC-DDoS.

You can download CICFlowMeter from the official source:

<https://www.unb.ca/cic/research/applications.html>

Using this tool ensures that your dataset follows the same feature structure that our models are trained on, resulting in more accurate analysis.

2. Using Existing Traffic Data (Wireshark or Other Tools)

If you already have recorded network traffic using **Wireshark** or any other packet capturing tool, you can still use it with our system.

CICFlowMeter supports **offline mode**, which allows you to convert existing traffic files into the required format. You can upload your .pcap or .pcapng files into CICFlowMeter, and it will generate a CSV file containing standardized flow-based features.

This process helps convert traffic captured from different tools into an **official and internationally accepted format**, making it compatible with our DDoS Analyzer models.

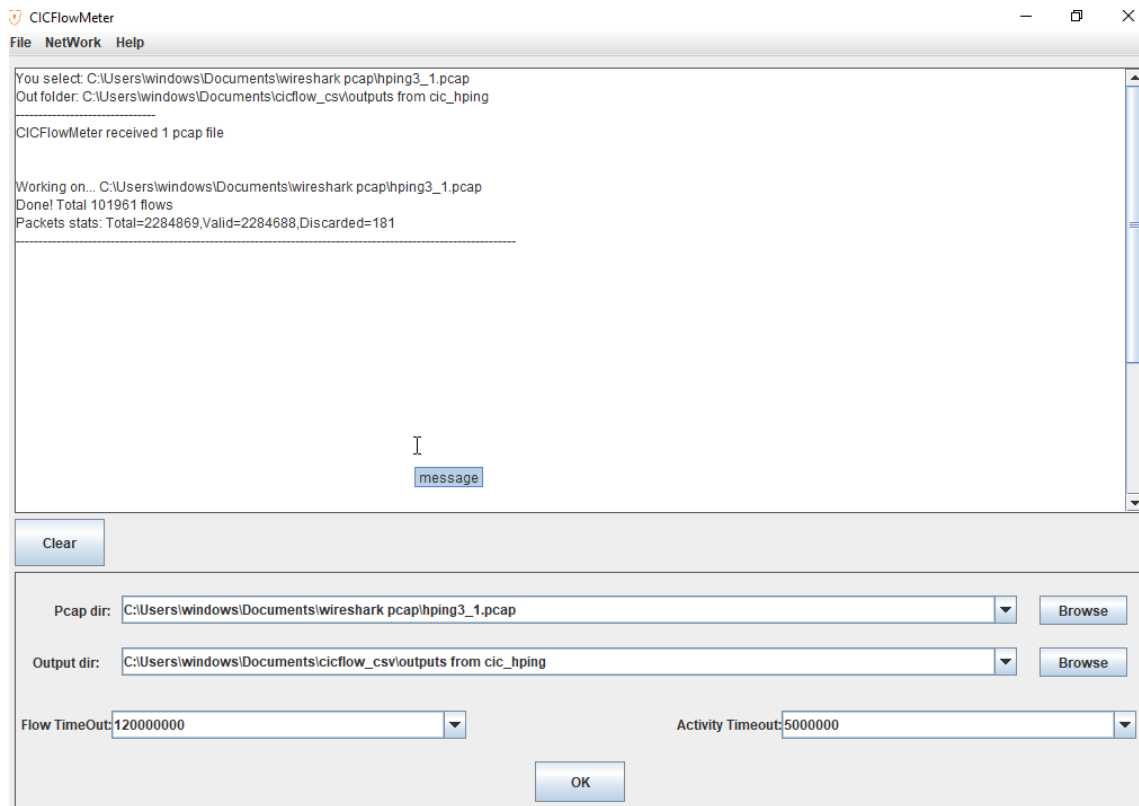


Figure showing CICFlowMeter offline mode converting a PCAP file into a CSV network traffic dataset.

3. Supported File Format

Only **CSV files** are supported for dataset upload.

Your CSV file should be generated directly from CICFlowMeter or converted using its offline mode. Uploading datasets that are not formatted through CICFlowMeter may lead to missing features, inconsistent data, or reduced detection accuracy.

4. Important Note on Results and Accuracy

If your dataset is not recorded using CICFlowMeter or not converted into its official format, the system may still process it, but the results might not be fully accurate or reliable.

To achieve optimal performance and correct classification between normal and DDoS traffic, **using CICFlowMeter is strongly recommended.**